3.2 The cardiac cycle

The cardiac cycle is the sequence of mechanical and electrical events that repeats with every heartbeat. Each cycle consists of a period of ventricular contraction, called systole, followed by a period of relaxation, called diastole. At a resting heart rate of 75 beats per minute, one cycle lasts about 0.8 seconds.

During isovolumetric contraction, all four valves are closed and ventricular pressure rises steeply without any change in volume. Once left ventricular pressure exceeds aortic pressure, about 80 mmHg, the aortic valve opens and ejection begins.

3.3 Stroke volume and cardiac output

Stroke volume is the difference between end-diastolic volume and end-systolic volume. In a healthy adult at rest, end-diastolic volume is about 120 mL and end-systolic volume is about 50 mL, giving a stroke volume of roughly 70 mL. Cardiac output is the product of stroke volume and heart rate.[1]

Preload, afterload and contractility each change stroke volume. Increased preload stretches the myocardium and, by the Frank-Starling mechanism, increases the force of contraction. Increased afterload, unlike increased preload, reduces stroke volume unless contractility also rises.

Sympathetic stimulation raises both heart rate and contractility, whereas parasympathetic stimulation through the vagus nerve lowers heart rate but has little direct effect on ventricular contractility.

3.4 Heart sounds

The first heart sound, S1, is produced by closure of the atrioventricular valves at the start of systole. The second heart sound, S2, is produced by closure of the semilunar valves at the start of diastole. A third sound, S3, can be normal in children but in older adults often indicates heart failure.

Physiological splitting of S2 occurs during inspiration, when the pulmonary valve closes slightly after the aortic valve; splitting that does not vary with breathing is not physiological and suggests an atrial septal defect.

[1] Cardiac output is usually reported in litres per minute; 70 mL x 75 beats/min = 5.25 L/min.